

A Note on the Heisenberg Calculus

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Abstract

This note presents an exposition of the Heisenberg calculus on contact manifolds, based on the incomplete notes and lectures of Charles L. Epstein and Richard B. Melrose [1, 2]. It aims to organize the construction and supply details of the arguments, from parabolic compactification and quantization to the symbolic calculus and the construction of parametrices.

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1 Parabolic compactification and Heisenberg symbol spaces

1.1 Contact data and weights

Let X be a smooth compact manifold without boundary of dimension $2n + 1$, $n \geq 1$, and let $H \subset TX$ be a cooriented contact distribution. Choose a contact form θ with $H = \ker \theta$ and $\theta \wedge (d\theta)^n \neq 0$. The Reeb vector field T is determined by $\theta(T) = 1$, $d\theta(T, \cdot) = 0$. We have

$$TX = H \oplus \mathbb{R}T, \quad T^*X = \mathbb{R}\theta \oplus H^*,$$

where H^* on the right is realized as the covectors annihilating T . Explicitly, for $\xi \in T_z^*X$,

$$\xi = \tau\theta_z + \eta, \quad \tau = \xi(T_z), \quad \eta = \xi - \xi(T_z)\theta_z.$$

Write this covector as (τ, η) . The dilations on T^*X are

$$M_\lambda(\tau, \eta) = (\lambda^2\tau, \lambda\eta), \quad \lambda > 0. \quad (1.1)$$

The central variable has weight 2, and the horizontal variables have weight 1. Set

$$|\alpha|_H = 2\alpha_0 + |\alpha'|, \quad Q = 2n + 2.$$

Here Q is the homogeneous dimension: the Jacobian of M_λ is λ^Q . The orders m, m' in the definitions and formulas may be real; each expansion proceeds through $m, m - 1, m - 2, \dots$. When discussing a filtered algebra, we restrict to integer orders. This distinction matters: for classical symbols, $S_H^{m-j} \subset S_H^m$ when $j \in \mathbb{N}$, but the same inclusion need not hold when j is a positive noninteger.

Throughout, we fix θ and its Reeb field T and work directly on $T^*X = \mathbb{R}\theta \oplus H^*$. Full symbols are functions on T^*X ; principal symbols are functions on $T^*X \setminus 0$ homogeneous under (1.1). This splitting depends on the chosen contact form, not on H alone. Changes of this choice induce maps on the parabolic compactification, as described below.

1.2 Adjoining the boundary at infinity

Choose a metric on H^* and, away from the zero section, set

$$R(\tau, \eta) = (\tau^2 + |\eta|^4)^{1/4}, \quad \rho = R^{-1}, \quad \Sigma_H = \{\tau^2 + |\eta|^4 = 1\}. \quad (1.2)$$

Since $R(M_\lambda\sigma) = \lambda R(\sigma)$, each dilation orbit meets Σ_H exactly once. Coordinates near infinity are

$$(\rho, \hat{\tau}, \hat{\eta}) = (R^{-1}, \rho^2\tau, \rho\eta), \quad \hat{\tau}^2 + |\hat{\eta}|^4 = 1. \quad (1.3)$$

Definition 1.1 (Parabolic compactification). The space ${}^H T^*X$ is obtained from T^*X by adjoining a copy of Σ_H at $\rho = 0$. Its smooth structure is the usual one in the interior and the one given by (1.3) near the boundary. Thus

$$\partial({}^H T^*X) = \Sigma_H.$$

Extend ρ to a smooth positive function in the interior, equal to R^{-1} near the boundary. It is a boundary defining function: $\rho \geq 0$, $\rho^{-1}(0) = \Sigma_H$, and $d\rho \neq 0$ on the boundary.

Proposition 1.2. *This smooth structure is independent of the metric and splitting; it depends only on the line bundle $H^\perp \subset T^*X$.*

Proof. Changing the metric multiplies R by a smooth positive function homogeneous of degree 0. A fiberwise linear change preserving $H^\perp$ has the form

$$\tilde{\tau} = c(z)\tau + \ell_z(\eta), \quad \tilde{\eta} = A_z\eta, \quad c \neq 0, \quad A_z \text{ invertible.}$$

In collar coordinates, put $F = ((c\hat{\tau} + \rho\ell(\hat{\eta}))^2 + |A\hat{\eta}|^4)^{1/4}$. Near the boundary, F is smooth and positive, and

$$\tilde{\rho} = \rho/F, \quad \tilde{\tau} = (c\hat{\tau} + \rho\ell(\hat{\eta}))/F^2, \quad \tilde{\eta} = A\hat{\eta}/F.$$

Applying the same calculation to the inverse gives a diffeomorphism up to the boundary. At $\rho = 0$, the term ℓ disappears, so the boundary map is induced by $(\tau, \eta) \mapsto (c\tau, A_z\eta)$. $\square$

1.3 Symbol classes

Definition 1.3 (Classical Heisenberg symbols). The space of Heisenberg symbols of order m is

$$S_H^m(X) = \rho^{-m} C^\infty({}^H T^* X). \quad (1.4)$$

Thus $a \in C^\infty(T^*X)$ belongs to S_H^m when $\rho^m a$ extends smoothly to the entire parabolic boundary. The homogeneous symbol space is

$$\mathcal{H}^m(X) = \{p \in C^\infty(T^*X \setminus 0) : p(z, M_\lambda \sigma) = \lambda^m p(z, \sigma)\}.$$

Using the fixed splitting, we write p as $p(z, \tau, \eta)$.

Proposition 1.4 (Expansions and estimates). *Every $a \in S_H^m$ has an expansion*

$$a \sim \sum_{j \geq 0} a_{m-j}, \quad a_{m-j} \in \mathcal{H}^{m-j}. \quad (1.5)$$

A cutoff near the zero section is understood in each term. After N terms, the remainder belongs to S_H^{m-N} . In adapted variables,

$$|\partial_z^\gamma \partial_\tau^{\alpha_0} \partial_\eta^{\alpha'} a(z, \tau, \eta)| \leq C_{\gamma\alpha} \langle \sigma \rangle_H^{m-|\alpha|_H}, \quad \langle \sigma \rangle_H = (1 + \tau^2 + |\eta|^4)^{1/4}. \quad (1.6)$$

Conversely, a smooth function with an expansion (1.5) whose remainders satisfy (1.6) at every order, after arbitrary differentiation, belongs to S_H^m .

Proof. Write $a = \rho^{-m} u$. Taylor expansion in ρ gives $u \sim \sum_{j \geq 0} \rho^j u_j(z, \hat{\sigma})$; set $a_{m-j} = \rho^{-m+j} u_j$ away from the zero section. In collar coordinates, ∂_{η_k} is ρ times a smooth vector field tangent to the boundary, while ∂_τ is ρ^2 times such a field. This proves (1.6). For the converse, take a smooth Borel sum of the jets u_j using Lemma 5.4. Its difference from $\rho^m a$ vanishes to every order, together with all derivatives tangent to the boundary. Since $\partial_\rho = \rho^{-1}(\rho \partial_\rho)$, the difference also vanishes to every order after ordinary differentiation and therefore extends smoothly by 0. $\square$

Remark 1.5. The estimates (1.6) alone do not define the classical class (1.4). For example, $1/\log(2+R)$, smoothed near the zero section, satisfies order-0 estimates but has no Taylor expansion in ρ . Moreover,

$$S_H^{-\infty} := \bigcap_{j \geq 0} S_H^{-j} = \dot{C}^\infty({}^H T^* X) \quad (1.7)$$

is the fiberwise Schwartz class; the dot denotes infinite-order vanishing at the boundary.

2 Quantization and Heisenberg operators

2.1 The Darboux model and the Heisenberg group

In a Darboux chart write $z = (t, x, y)$, $x, y \in \mathbb{R}^n$, with

$$\theta = dt + \frac{1}{2}(x \cdot dy - y \cdot dx), \quad X_j = \partial_{x_j} + \frac{1}{2}y_j \partial_t, \quad Y_j = \partial_{y_j} - \frac{1}{2}x_j \partial_t, \quad T = \partial_t. \quad (2.1)$$

The only nonzero brackets are $[X_j, Y_k] = -\delta_{jk}T$. For a coordinate covector ξ , the adapted variables are

$$\tau = \xi_t, \quad \eta_x = \xi_x + \frac{1}{2}y\xi_t, \quad \eta_y = \xi_y - \frac{1}{2}x\xi_t. \quad (2.2)$$

These are the symbols of D_T, D_X, D_Y , where $D_V = V/i$. We use the group law

$$(t, x, y) \cdot (s, u, v) = (t + s + \frac{1}{2}(y \cdot u - x \cdot v), x + u, y + v). \quad (2.3)$$

The fields (2.1) are left-invariant; $z^{-1} = -z$; and Haar measure is Lebesgue measure. This is the opposite group law to the one used in the supplied source. At the same time, we use $w^{-1} \cdot z$ instead of $z \cdot w^{-1}$, so the phase is unchanged. Set $d = 2n + 1$. Direct substitution of (2.2) gives

$$\langle z - w, \xi \rangle = \langle w^{-1} \cdot z, \sigma \rangle, \quad \sigma = (\tau, \eta). \quad (2.4)$$

In this identity, the fiberwise change of variables may be taken at z or at $(z + w)/2$; its Jacobian is 1.

2.2 Weyl and left quantization

Choose a smooth cutoff $\chi(z, w)$ equal to 1 near the diagonal and supported within the chart. For $a = a(z, \sigma)$, define

$$\text{Op}_H^W(a)f(z) = (2\pi)^{-d} \text{Os} \int e^{i\langle w^{-1} \cdot z, \sigma \rangle} \chi(z, w) a((z + w)/2, \sigma) f(w) dw d\sigma, \quad (2.5)$$

$$\text{Op}_H^L(a)f(z) = (2\pi)^{-d} \text{Os} \int e^{i\langle w^{-1} \cdot z, \sigma \rangle} \chi(z, w) a(z, \sigma) f(w) dw d\sigma. \quad (2.6)$$

The oscillatory integrals are interpreted by inserting $\psi(\varepsilon\sigma)$, with $\psi \in C_c^\infty$ equal to 1 near 0, and taking the distributional limit as $\varepsilon \downarrow 0$. Integration by parts proves independence of the cutoff. By (2.4), (2.5) is precisely the source's Weyl quantization written in adapted fiber variables.

Put $k_a(z, v) = \mathcal{F}_{\sigma \rightarrow v}^{-1} a(z, \sigma)$, with normalization $\mathcal{F}^{-1} a = (2\pi)^{-d} \int e^{i\langle v, \sigma \rangle} a(\sigma) d\sigma$. Suppressing the cutoff, we have

$$\text{Op}_H^L(a)f(z) = \int k_a(z, v) f(z \cdot v^{-1}) dv. \quad (2.7)$$

Definition 2.1. An operator A belongs to $\Psi_H^m(X)$ if its kernel is smooth off the diagonal and, after localization in each Darboux chart, it is of the form $\text{Op}_H^W(a)$ modulo a smooth kernel, with $a \in S_H^m$. Write $\Psi^{-\infty}(X)$ for the operators with smooth kernels on $X \times X$.

Lemma 2.2 (Kernels and cutoffs). *Let $a \in S_H^m$, $\bullet \in \{L, W\}$, and $A_\chi = \text{Op}_{H,\chi}^\bullet(a)$. Write $\Delta = \{(z, z) : z \in X\}$.*

(i) *Off-diagonal smoothness and cutoff independence:*

$$K_{A_\chi} \in C^\infty((X \times X) \setminus \Delta), \quad A_{\chi_1} - A_{\chi_2} \in \Psi^{-\infty}(X).$$

Here $\chi_1 = \chi_2 = 1$ near Δ .

(ii) **Continuity:**

$$A_\chi : C^\infty(X) \longrightarrow C^\infty(X), \quad A_\chi : \mathcal{D}'(X) \longrightarrow \mathcal{D}'(X)$$

are continuous.

(iii) **Schwartz symbols and smooth kernels:**

$$a \in S_H^{-\infty} \implies A_\chi \in \Psi^{-\infty}(X), \quad k(z, v) \in C_c^\infty \implies \mathcal{F}_{v \rightarrow \sigma} k \in S_H^{-\infty}.$$

The last assertion applies to a smooth kernel localized in a chart and written in the variables (z, v) , $v = w^{-1} \cdot z$.

Proof. (i) For $v \neq 0$, set

$$L_v = \frac{v \cdot \partial_\sigma}{i|v|^2}, \quad L_v e^{i\langle v, \sigma \rangle} = e^{i\langle v, \sigma \rangle}.$$

For $k_a = \mathcal{F}^{-1}a$, integration by parts gives

$$\partial_z^\gamma \partial_v^\beta k_a(z, v) = (2\pi)^{-d} \int e^{i\langle v, \sigma \rangle} (L_v^t)^N [(i\sigma)^\beta \partial_z^\gamma a(z, \sigma)] d\sigma.$$

On compact sets with $v \neq 0$, (1.6) implies

$$\left| (L_v^t)^N [(i\sigma)^\beta \partial_z^\gamma a] \right| \leq C \langle \sigma \rangle_H^{m+2|\beta|-N}, \quad \int \langle \sigma \rangle_H^{-s} d\sigma < \infty \quad (s > Q).$$

Choose $N > m + 2|\beta| + Q$. Thus k_a is smooth for $v \neq 0$, and

$$K_{A_\chi}(z, w) = \chi(z, w) \begin{cases} k_a(z, w^{-1} \cdot z), & \bullet = L, \\ k_a((z+w)/2, w^{-1} \cdot z), & \bullet = W \end{cases}$$

is smooth off Δ . Multiplication by $\chi_1 - \chi_2$, which vanishes near Δ , gives a smooth kernel everywhere.

(ii) Put $w = z \cdot v^{-1}$ and

$$Z_L(z, v) = z, \quad Z_W(z, v) = (z+w)/2, \quad b_f(z, v, \sigma) = \chi(z, w) a(Z_\bullet(z, v), \sigma) f(w).$$

The v -support is compact. Using $(1 - \Delta_v)^N e^{i\langle v, \sigma \rangle} = \langle \sigma \rangle^{2N} e^{i\langle v, \sigma \rangle}$, we obtain

$$A_\chi f(z) = (2\pi)^{-d} \int e^{i\langle v, \sigma \rangle} \langle \sigma \rangle^{-2N} (1 - \Delta_v)^N b_f(z, v, \sigma) dv d\sigma.$$

Here $\langle \sigma \rangle = (1 + |\sigma|^2)^{1/2}$. On each localized chart, for $2N > d + \max(m, 0)$,

$$\|A_\chi f\|_{C^k} \leq C_{k,N,a} \|f\|_{C^{k+2N}} \int \langle \sigma \rangle^{-2N} \langle \sigma \rangle^m d\sigma \leq C'_{k,N,a} \|f\|_{C^{k+2N}}.$$

This proves continuity on C^∞ . The same estimate applies to the transposed kernel $K_{A_\chi^t}(z, w) = K_{A_\chi}(w, z)$. The continuous extension to distributions is therefore defined by

$$\langle A_\chi u, \varphi \rangle = \langle u, A_\chi^t \varphi \rangle, \quad u \in \mathcal{D}'(X), \quad \varphi \in C^\infty(X).$$

(iii) Fourier transformation preserves the Schwartz space:

$$a(z, \cdot) \in \mathcal{S} \implies k_a(z, \cdot) \in \mathcal{S} \implies K_{A_\chi} \in C^\infty.$$

Conversely, for $k \in C_c^\infty$ and every N, α, γ ,

$$\langle \sigma \rangle^{2N} \partial_z^\gamma \partial_\sigma^\alpha \widehat{k}(z, \sigma) = \mathcal{F}_{v \rightarrow \sigma} [(1 - \Delta_v)^N ((-iv)^\alpha \partial_z^\gamma k)]$$

is bounded by the L^1 norm of the compactly supported expression in brackets. Hence $\widehat{k}$ is fiberwise Schwartz, with smooth dependence on z . $\square$

Proposition 2.3 (Changes of quantization and coordinates). *Weyl quantization may be replaced by left quantization in Definition 2.1. The two full symbols differ by a symbol of order $m - 1$. The class Ψ_H^m is independent of the contact chart, and its leading term transforms by the map induced on the parabolic boundary.*

Proof. Change of quantization. Put $Z = (T, X, Y)$ and $V_v = \sum_j v_j Z_j$. Then

$$w = zv^{-1} \implies \frac{z+w}{2} = z \cdot (-v/2) = z \exp(-v/2).$$

For $|\alpha| = k$, let Z_{sym}^α be the average of all orderings with multiplicities α , and set $D_{\text{sym}}^\alpha = i^{-k} Z_{\text{sym}}^\alpha$. Thus

$$V_v^k = \sum_{|\alpha|=k} \frac{k!}{\alpha!} v^\alpha Z_{\text{sym}}^\alpha, \quad \left. \frac{d^k}{ds^k} a_W(z \exp(-sv/2), \sigma) \right|_{s=0} = (-1/2)^k V_v^k a_W(z, \sigma).$$

Taylor expansion with integral remainder gives, for every $N \geq 1$,

$$a_W(z \exp(-v/2), \sigma) = \sum_{|\alpha|_H < N} \frac{(-1/2)^{|\alpha|} v^\alpha}{\alpha!} Z_{\text{sym}}^\alpha a_W(z, \sigma) + \mathcal{R}_N(z, v, \sigma).$$

By Lemma 5.1(i), independently of the present proposition,

$$\begin{aligned} \mathcal{F}(v^\alpha k) &= i^{|\alpha|} \partial_\sigma^\alpha \mathcal{F}k, \\ (-1/2)^{|\alpha|} i^{|\alpha|} Z_{\text{sym}}^\alpha &= 2^{-|\alpha|} D_{\text{sym}}^\alpha, \\ a_L - \sum_{|\alpha|_H < N} \frac{2^{-|\alpha|}}{\alpha!} \partial_\sigma^\alpha D_{\text{sym}}^\alpha a_W &\in S_H^{m-N}. \end{aligned}$$

Consequently,

$$a_L \sim \sum_{\alpha \in \mathbb{N}^{2n+1}} \frac{2^{-|\alpha|}}{\alpha!} \partial_\sigma^\alpha D_{\text{sym}}^\alpha a_W. \quad (2.8)$$

$$\partial_\sigma^\alpha D_{\text{sym}}^\alpha : S_H^m \longrightarrow S_H^{m-|\alpha|_H}, \quad \alpha \neq 0 \implies |\alpha|_H \geq 1 \implies a_L - a_W \in S_H^{m-1}.$$

For the inverse, denote the order- j correction in (2.8) by

$$P_0 = \text{Id}, \quad P_j = \sum_{|\alpha|_H=j} \frac{2^{-|\alpha|}}{\alpha!} \partial_\sigma^\alpha D_{\text{sym}}^\alpha.$$

The recursion

$$Q_0 = \text{Id}, \quad Q_j = - \sum_{k=1}^j Q_{j-k} P_k, \quad Q_j : S_H^m \longrightarrow S_H^{m-j},$$

followed by Borel summation gives $a_W \sim \sum_{j \geq 0} Q_j a_L$. Hence both quantizations define the same operator class modulo smoothing kernels.

Change of contact coordinates. Let $\phi^* \theta' = f \theta$, and use the fixed Reeb splittings. Set

$$L_z = d_H \phi_z, \quad L_z(v_0 T_z + h) = f(z) v_0 T'_{\phi(z)} + d\phi_z(h), \quad h \in H_z.$$

Writing $A_z = d\phi_z|_{H_z}$, we have

$$L_z(v_0, h) = (f(z) v_0, A_z h), \quad L_z \delta_\varepsilon = \delta_\varepsilon L_z, \quad A_z^* \omega_0 = f(z) \omega_0,$$

where $\omega_0((x, y), (\tilde{x}, \tilde{y})) = x \cdot \tilde{y} - y \cdot \tilde{x}$. For transitions preserving the fixed contact form, $f = 1$, $d\phi_z(T_z) = T'_{\phi(z)}$, and $L_z = d\phi_z$.

Put $F_z(v) = \phi(z)^{-1} \cdot \phi(z \cdot v)$. Write $O_H(r)$ for a smooth remainder whose Taylor monomials have weight at least r . Initially,

$$F'_z(v_0, h) = A_z h + O_H(2), \quad F_z^0(v_0, h) = f(z)v_0 + \frac{1}{2}B_z(h, h) + O_H(3),$$

with B_z symmetric. The horizontal terms of weight 1 in $F_z^*(dv_0 + \frac{1}{2}\omega_0(h, dh)) = f(z \cdot v)(dv_0 + \frac{1}{2}\omega_0(h, dh))$ give

$$\frac{1}{2}d_h(B_z(h, h)) + \frac{1}{2}\omega_0(A_z h, A_z dh) = \frac{1}{2}f(z)\omega_0(h, dh).$$

Since $A_z^* \omega_0 = f(z)\omega_0$,

$$d_h(B_z(h, h)) = 0 \implies B_z = 0 \implies \begin{cases} F_z^0 - f(z)v_0 = O_H(3), \\ F'_z - A_z h = O_H(2). \end{cases}$$

Equivalently,

$$\delta_\varepsilon^{-1} F_z(\delta_\varepsilon v) = d_H \phi_z(v) + \varepsilon E(z, v, \varepsilon), \quad (2.9)$$

with E smooth down to $\varepsilon = 0$ on compact sets.

For $w = zv^{-1}$, the transformed kernel phase is determined by

$$\phi(w)^{-1} \phi(z) = -F_z(-v) = L_z v + \mathcal{E}_z(v), \quad \mathcal{E}_z^0 = O_H(3), \quad \mathcal{E}'_z = O_H(2).$$

Change frequency variables by $\sigma = L_z^t \sigma'$. Then

$$\langle \phi(w)^{-1} \phi(z), \sigma' \rangle = \langle v, \sigma \rangle + \langle \mathcal{E}_z(v), (L_z^t)^{-1} \sigma \rangle.$$

For operators on functions, the Jacobian amplitude is

$$J(z, v) = \frac{|\det d\phi_{zv^{-1}}|}{|\det L_z|}, \quad J(z, 0) = 1, \quad J(z, v) - 1 = O_H(1).$$

For half-densities, the numerator is instead $|\det d\phi_z|^{1/2} |\det d\phi_{zv^{-1}}|^{1/2}$, with the same value 1 at $v = 0$. Every nonconstant phase correction satisfies

term	weight in v	frequency weight	order drop
$v^\alpha \eta_j$	$ \alpha _H \geq 2$	1	$ \alpha _H - 1 \geq 1$
$v^\alpha \tau$	$ \alpha _H \geq 3$	2	$ \alpha _H - 2 \geq 1$.

Thus Lemma 5.1, applied to the phase exponential and J , gives for the transformed left symbol

$$a_\phi(z, \sigma) - a'(\phi(z), (L_z^t)^{-1} \sigma) \in S_H^{m-1}, \quad a_\phi \in S_H^m.$$

In particular, the leading terms satisfy

$$p_\phi(z, \sigma) = p'(\phi(z), (L_z^t)^{-1} \sigma).$$

By Proposition 1.2, this is precisely the map induced on the parabolic boundary. Applying the same argument to ϕ^{-1} proves independence of the contact chart. $\square$

3 Symbol maps and the two parabolic hemispheres

3.1 The principal symbol and the exact sequence

With $\rho = R^{-1}$ near the boundary, define

$$\beta_m : S_H^m \rightarrow C^\infty(\Sigma_H), \quad \beta_m(a) = (\rho^m a)|_{\Sigma_H}.$$

Every $\sigma \neq 0$ has the unique form

$$\sigma = M_r \hat{\sigma}, \quad r = R(\sigma) > 0, \quad \hat{\sigma} \in \Sigma_H.$$

Thus restriction to Σ_H has the inverse

$$\mathcal{E}_m : C^\infty(\Sigma_H) \longrightarrow \mathcal{H}^m, \quad (\mathcal{E}_m f)(z, M_r \hat{\sigma}) = r^m f(z, \hat{\sigma}). \quad (3.1)$$

Definition 3.1 (Principal symbol of a full symbol). The principal symbol of $a \in S_H^m$ is the homogeneous extension of its boundary value:

$$p_a = \mathcal{E}_m \beta_m(a), \quad p_a(z, \sigma) = R(\sigma)^m \beta_m(a)(z, \hat{\sigma}). \quad (3.2)$$

Smoothness and homogeneity follow directly from (3.1). Moreover,

$$b \in S_H^{m-1} \implies \beta_m(b) = \rho(\rho^{m-1}b)|_{\Sigma_H} = 0 \implies p_{a+b} = p_a.$$

The familiar limit formula is a consequence, not a separate definition: writing $u = \rho^m a$ in collar coordinates gives

$$\lambda^{-m} a(z, M_\lambda \sigma) = R(\sigma)^m u(z, (\lambda R(\sigma))^{-1}, \hat{\sigma}) \longrightarrow p_a(z, \sigma). \quad (3.3)$$

Theorem 3.2 (The principal symbol). For a local full symbol a of A , the assignment $\sigma_m^H(A) = \mathcal{E}_m \beta_m(a)$ defines $\sigma_m^H : \Psi_H^m \rightarrow \mathcal{H}^m$ and the exact sequence

$$0 \longrightarrow \Psi_H^{m-1}(X) \longrightarrow \Psi_H^m(X) \xrightarrow{\sigma_m^H} \mathcal{H}^m(X) \longrightarrow 0. \quad (3.4)$$

Moreover, $\bigcap_{j \geq 0} \Psi_H^{-j} = \Psi^{-\infty}(X)$.

Proof. (i) Well-definedness. For two local full symbols $a, \tilde{a}$ of the same operator, Lemma 2.2 and Proposition 2.3 give

$$a - \tilde{a} \in \begin{cases} S_H^{-\infty}, & \text{same quantization, different cutoffs,} \\ S_H^{m-1}, & \text{Weyl versus left quantization.} \end{cases}$$

Since $S_H^{-\infty} \subset S_H^{m-1}$,

$$\beta_m(a - \tilde{a}) = \rho(\rho^{m-1}(a - \tilde{a}))|_{\Sigma_H} = 0 \implies \mathcal{E}_m \beta_m(a) = \mathcal{E}_m \beta_m(\tilde{a}).$$

The coordinate transformation law in Proposition 2.3 identifies these local expressions on overlaps, giving $\sigma_m^H(A) \in \mathcal{H}^m(X)$.

(ii) *Kernel.* Set $u = \rho^m a \in C^\infty({}^H T^* X)$. Hadamard's lemma gives

$$\begin{aligned} \sigma_m^H(A) = 0 &\iff \beta_m(a) = 0 \iff u|_{\Sigma_H} = 0 \\ &\iff u = \rho u_1, \quad u_1 \in C^\infty({}^H T^* X) \\ &\iff a = \rho^{-(m-1)} u_1 \in S_H^{m-1} \iff A \in \Psi_H^{m-1}. \end{aligned}$$

These equivalences hold in every localized chart; hence $\ker \sigma_m^H = \Psi_H^{m-1}(X)$.

(iii) *Surjectivity*. Given $p \in \mathcal{H}^m(X)$, choose

$$\chi(R) = \begin{cases} 0, & R \leq 1, \\ 1, & R \geq 2, \end{cases} \quad \sum_j \varphi_j = 1, \quad \text{supp } \varphi_j \Subset U_j,$$

with χ smooth and $\{U_j\}$ a finite Darboux cover. Localize each quantization inside $U_j \times U_j$ and set

$$a_j = \chi(R)\varphi_j p \in S_H^m, \quad A = \sum_j \text{Op}_{H,j}^W(a_j) \in \Psi_H^m(X).$$

The boundary values satisfy

$$\beta_m(a_j) = \varphi_j p|_{\Sigma_H}, \quad \sigma_m^H(A) = \sum_j \mathcal{E}_m \beta_m(a_j) = \sum_j \varphi_j p = p.$$

Finally, for a fixed local full symbol a of A , Fourier inversion and Lemma 2.2 yield

$$\begin{aligned} A \in \bigcap_{j \geq 0} \Psi_H^{-j} &\iff a \in \bigcap_{j \geq 0} S_H^{-j} \text{ in every chart} \\ &\iff a \text{ is fiberwise Schwartz in every chart} \\ &\iff K_A \in C^\infty(X \times X) \iff A \in \Psi^{-\infty}(X). \end{aligned}$$

□

3.2 Hemispheres and quadratic compactification

Set

$$\Sigma_H^\pm = \{\hat{\sigma} \in \Sigma_H : \pm \hat{\tau} \geq 0\}, \quad E = \Sigma_H^+ \cap \Sigma_H^- = \{\hat{\tau} = 0, |\hat{\eta}| = 1\}.$$

The two affine hyperplanes $\mathcal{P}_\pm = \{\tau = \pm 1\} \cong H^*$ are identified with the *interiors* of the hemispheres by

$$\eta \mapsto \left(\frac{\pm 1}{(1 + |\eta|^4)^{1/2}}, \frac{\eta}{(1 + |\eta|^4)^{1/4}} \right). \quad (3.5)$$

With $r = |\eta|$ and $\omega = \eta/r$, a normal coordinate to the equator within a hemisphere is $s = r^{-2}$, not r^{-1} , since

$$|\hat{\tau}| = \frac{s}{\sqrt{1 + s^2}}, \quad \hat{\eta} = (1 + s^2)^{-1/4} \omega.$$

This is the *quadratic compactification* of $\mathcal{P}_\pm$. Its smooth structure differs from that of the usual radial compactification when the identity map on the interior is kept fixed.

Definition 3.3 (Hemisphere and equatorial symbol maps). For $p = \sigma_m^H(A)$, set

$$\sigma_m^H(A)(\pm)(z, \eta) = p_\pm(z, \eta) := p(z, \pm 1, \eta), \quad (3.6)$$

$$\sigma_{\text{eq},m}(A)(z, \eta) = p(z, 0, \eta), \quad \eta \neq 0. \quad (3.7)$$

These are restrictions of the *homogeneous symbol*. Pulling back $\beta_m(a)$ by (3.5) gives $(1 + |\eta|^4)^{-m/4} p_\pm$, not $p_\pm$ itself.

Definition 3.4 (Classical isotropic symbols). On $\mathbb{R}^{2n}$, a smooth function f belongs to S_{iso}^m if $r^{-m}f(r\omega)$ is smooth in (r^{-1}, ω) down to $r^{-1} = 0$. Equivalently,

$$f(r\omega) \sim \sum_{j \geq 0} r^{m-j} f_j(\omega), \quad |\partial_\eta^\alpha f(\eta)| \leq C_\alpha \langle \eta \rangle^{m-|\alpha|},$$

with corresponding estimates for all remainders. The notation $S_{\text{iso}}^m(\mathcal{P}_\pm)$ requires smooth dependence on z , with estimates locally uniform in z .

Proposition 3.5 (Matching all jets at the equator). *The map $p \mapsto (p_+, p_-)$ from $\mathcal{H}^m$ to $S_{\text{iso}}^m(\mathcal{P}_+) \times S_{\text{iso}}^m(\mathcal{P}_-)$ is injective. Its image consists precisely of the pairs with expansions*

$$\boxed{p_{\pm}(z, r\omega) \sim \sum_{k \geq 0} (\pm 1)^k r^{m-2k} p_k(z, \omega),} \quad (3.8)$$

for the same sequence p_k , with remainder in S_{iso}^{m-2K} after K terms, including all differentiated estimates. In this case, $p(z, 0, r\omega) = r^m p_0(z, \omega)$.

Proof. Homogeneity gives the exact identity

$$p(z, \pm 1, r\omega) = r^m p(z, \pm r^{-2}, \omega). \quad (3.9)$$

Taylor expansion at $\tau = 0$ yields (3.8), with $p_k = (k!)^{-1} \partial_{\tau}^k p(z, 0, \omega)$. Differentiating the identity gives all remainder estimates. Conversely, given a matched pair, define

$$p(z, \tau, \eta) = |\tau|^{m/2} p_{\text{sgn } \tau}(z, |\tau|^{-1/2} \eta), \quad \tau \neq 0. \quad (3.10)$$

Near $(0, \omega)$, the two sides have the same expansion $\sum_k \tau^k p_k(z, \omega)$. The remainder is $O(|\tau|^{K-j})$ after j derivatives in τ , including horizontal and base derivatives. Subtracting a Borel sum of the jets leaves a function that extends by 0 to a smooth function flat at $\tau = 0$. Thus p extends smoothly and homogeneously. Uniqueness follows from the density of $\{\tau \neq 0\}$. $\square$

Remark 3.6. With the convention (1.1), the isotropic order of $p_{\pm}$ is m , not $2m$; the supplied document also emphasizes this point. The two restrictions are not arbitrary isotropic symbols: their expansions decrease in steps of 2, with the sign $(-1)^k$ relating the two sides. Matching only the leading term at the equator does not suffice for smooth gluing.

4 Isotropic algebras and gluing the principal-symbol product

4.1 The exact Weyl product

On each fiber write $\eta = (q, p) \in \mathbb{R}^n \times \mathbb{R}^n$ and set

$$\omega((q, p), (q', p')) = q \cdot p' - p \cdot q'.$$

The sign convention is chosen so that the symbol commutator of D_X, D_Y is $i\tau$. The fibers $\mathcal{P}_{\pm}$ carry the symplectic structures $\pm\omega$. For $\varepsilon \in \{+1, -1\}$, model quantization is

$$q_{\varepsilon}^W(f)u(q) = (2\pi)^{-n} \text{Os} \int e^{i\varepsilon(q-q') \cdot p} f((q+q')/2, p) u(q') dq' dp. \quad (4.1)$$

For Schwartz functions f, g , composing the kernels and applying Fourier inversion gives

$$\begin{aligned} q_{\varepsilon}^W(f)q_{\varepsilon}^W(g) &= q_{\varepsilon}^W(f \#_{\varepsilon} g), \\ (f \#_{\varepsilon} g)(\eta) &= \pi^{-2n} \text{Os} \int e^{2i\varepsilon\omega(u, v)} f(\eta + u) g(\eta + v) du dv. \end{aligned} \quad (4.2)$$

To check the constant, in the composed kernel set the intermediate point equal to $q + u_q + v_q$ and the difference of the two outer points equal to $2(u_q - v_q)$; translate both frequencies by p . The Jacobian is 2^{2n} , the phase is $2\varepsilon\omega(u, v)$, and $(2\pi)^{-2n} 2^{2n} = \pi^{-2n}$. Define

$$\begin{aligned} \mathcal{P} &= \sum_j (\partial_{q_j} \partial_{p'_j} - \partial_{p_j} \partial_{q'_j}), \\ B_k(f, g) &= \frac{1}{k!} \left(\frac{i}{2}\right)^k \mathcal{P}^k [f(q, p) g(q', p')] |_{(q', p') = (q, p)}. \end{aligned} \quad (4.3)$$

In particular, $B_0(f, g) = fg$ and $B_1(f, g) = \frac{i}{2} \sum_j (f_{q_j} g_{p_j} - f_{p_j} g_{q_j})$.

Lemma 4.1 (The isotropic calculus). *The product (4.2) extends continuously to an associative product $\#_\varepsilon : S_{\text{iso}}^m \times S_{\text{iso}}^{m'} \rightarrow S_{\text{iso}}^{m+m'}$ with unit 1. It satisfies*

$$f \#_\varepsilon g \sim \sum_{k \geq 0} \varepsilon^k B_k(f, g), \quad f \#_\varepsilon g - \sum_{k < K} \varepsilon^k B_k(f, g) \in S_{\text{iso}}^{m+m'-2K}. \quad (4.4)$$

The Schwartz space $\mathcal{S}(\mathbb{R}^{2n})$ is a two-sided ideal, and complex conjugation reverses the order of the product.

Proof. Put $R = \langle \eta \rangle$ and split the integral into $|(u, v)| < cR$, with c small, and its complement. The phase $\Phi = 2\varepsilon\omega(u, v)$ is a nondegenerate quadratic form. The operator

$$L = \frac{1 - i\nabla\Phi \cdot \nabla_{u,v}}{1 + |\nabla\Phi|^2} \quad \text{satisfies} \quad Le^{i\Phi} = e^{i\Phi}.$$

Outside the near region,

$$|\nabla\Phi| = 2|(u, v)| \geq 2cR.$$

Repeated integration by parts using L makes this contribution rapidly decreasing in R . If one argument $\eta + u$ or $\eta + v$ is small, the symbol derivatives still grow at most polynomially in $R + |(u, v)|$; increasing the number of integrations by parts absorbs this growth.

In the near region, with $0 < c < 1$,

$$\begin{aligned} (1 - c)R &\leq \langle \eta + u \rangle \leq (1 + c)R, \\ (1 - c)R &\leq \langle \eta + v \rangle \leq (1 + c)R. \end{aligned}$$

Set $u = RU$, $v = RV$. The phase parameter is R^2 , and the amplitude divided by $R^{m+m'}$ has uniformly bounded derivatives of every order. Quadratic stationary phase follows from Gaussian Fourier inversion: Taylor-expand the amplitude, integrate monomials using derivatives of the Gaussian, and use L for the remainder away from the stationary point. Each pair of derivatives contributes $i\varepsilon/2$ and a factor R^{-2} . After K steps, the derivative of order α of the remainder is bounded by $C_\alpha R^{m+m'-2K-|\alpha|}$. This proves (4.4).

Substituting the classical expansions of f, g shows that the product is classical. Associativity holds for Schwartz symbols by composition of kernels. For general symbols, insert frequency cutoffs and remove them in symbol classes of slightly larger order; the estimates just proved pass the identity to the limit. The unit and conjugation properties follow in the same way. If one factor is Schwartz, take its order arbitrarily negative in the product estimate to obtain the ideal property. $\square$

Remark 4.2. Formula (4.2) defines the *exact* product, whereas (4.4) is only an asymptotic expansion and need not converge. The exact product uses only ω and Liouville measure, so it is invariant under linear symplectic changes of variables. Consequently, the products $\#_\pm$ patch into bundles of algebras over X , without requiring a global Lagrangian splitting.

4.2 Gluing the two products across the equator

Theorem 4.3 (The homogeneous-symbol product). *For $p \in \mathcal{H}^m$ and $q \in \mathcal{H}^{m'}$, the pair $(p_+ \#_+ q_+, p_- \#_- q_-)$ satisfies the matching condition at order $m + m'$. Hence there is a unique $p \star q \in \mathcal{H}^{m+m'}$ with these restrictions. The product $\star$ is associative, has unit 1, and satisfies*

$$(p \star q)(z, 0, \eta) = p(z, 0, \eta)q(z, 0, \eta). \quad (4.5)$$

Proof. (i) Homogeneous coefficients. Suppress z and write $\eta = r\omega$, $r > 0$, $|\omega| = 1$. By (3.8), for $\varepsilon \in \{+1, -1\}$,

$$p_\varepsilon(r\omega) \sim \sum_{a \geq 0} \varepsilon^a r^{m-2a} p_a(\omega), \quad q_\varepsilon(r\omega) \sim \sum_{b \geq 0} \varepsilon^b r^{m'-2b} q_b(\omega).$$

Define, for $\eta \neq 0$,

$$P_a(\eta) = r^{m-2a} p_a(\omega), \quad Q_b(\eta) = r^{m'-2b} q_b(\omega).$$

Each term in $B_k(P_a, Q_b)$ differentiates each factor k times; hence

$$\begin{aligned} \deg_\eta B_k(P_a, Q_b) &= (m - 2a - k) + (m' - 2b - k) \\ &= m + m' - 2(a + b + k), \\ \varepsilon^k B_k(\varepsilon^a P_a, \varepsilon^b Q_b) &= \varepsilon^{a+b+k} B_k(P_a, Q_b). \end{aligned}$$

Thus the terms with $a + b + k = \ell$ have the same degree and sign. Set

$$C_\ell(\eta) = \sum_{a+b+k=\ell} B_k(P_a, Q_b)(\eta), \quad c_\ell = C_\ell|_{|\eta|=1}. \quad (4.6)$$

In particular,

$$C_\ell(r\omega) = r^{m+m'-2\ell} c_\ell(\omega), \quad c_\ell \text{ is independent of } \varepsilon.$$

(ii) *Remainders and matching.* Let $\chi(\eta) = 0$ near 0 and $\chi(\eta) = 1$ for $|\eta| \geq 2$. For every $N \geq 1$,

$$p_\varepsilon - \sum_{a < N} \varepsilon^a \chi P_a \in S_{\text{iso}}^{m-2N}, \quad q_\varepsilon - \sum_{b < N} \varepsilon^b \chi Q_b \in S_{\text{iso}}^{m'-2N}.$$

Put $h_\varepsilon = p_\varepsilon \#_\varepsilon q_\varepsilon$. The finite expansion (4.4) gives

$$h_\varepsilon = \sum_{k < N} \varepsilon^k B_k(p_\varepsilon, q_\varepsilon) \pmod{S_{\text{iso}}^{m+m'-2N}}.$$

Substitute the two finite symbol expansions. Terms containing a remainder, or satisfying $a + b + k \geq N$, lie in $S_{\text{iso}}^{m+m'-2N}$; derivatives of χ give compactly supported terms. Consequently,

$$h_\varepsilon - \sum_{\ell < N} \varepsilon^\ell \chi C_\ell \in S_{\text{iso}}^{m+m'-2N}.$$

More explicitly, for $|\eta| \geq 2$ and every α, γ ,

$$\left| \partial_z^\gamma \partial_\eta^\alpha \left(h_\varepsilon(z, \eta) - \sum_{\ell < N} \varepsilon^\ell C_\ell(z, \eta) \right) \right| \leq C_{N, \alpha, \gamma} \langle \eta \rangle^{m+m'-2N-|\alpha|}.$$

Therefore

$$h_\pm(z, r\omega) \sim \sum_{\ell \geq 0} (\pm 1)^\ell r^{m+m'-2\ell} c_\ell(z, \omega).$$

Proposition 3.5 now yields

$$\exists! p \star q \in \mathcal{H}^{m+m'} : \quad (p \star q)_\pm = h_\pm.$$

Here $h_\pm$ are the *exact* products; their expansions are used only to verify matching at the equator.

(iii) *Algebraic properties and the equator.* For a third homogeneous symbol s ,

$$\begin{aligned} ((p \star q) \star s)_\varepsilon &= (p_\varepsilon \#_\varepsilon q_\varepsilon) \#_\varepsilon s_\varepsilon \\ &= p_\varepsilon \#_\varepsilon (q_\varepsilon \#_\varepsilon s_\varepsilon) = (p \star (q \star s))_\varepsilon, \\ (1 \star p)_\varepsilon &= 1 \#_\varepsilon p_\varepsilon = p_\varepsilon = p_\varepsilon \#_\varepsilon 1 = (p \star 1)_\varepsilon. \end{aligned}$$

Injectivity of restriction gives associativity and the unit. Finally,

$$\begin{aligned} c_0(\omega) &= B_0(P_0, Q_0)(\omega) = p_0(\omega)q_0(\omega), \\ (p \star q)(z, 0, r\omega) &= r^{m+m'} c_0(z, \omega) \\ &= (r^m p_0(z, \omega)) (r^{m'} q_0(z, \omega)) \\ &= p(z, 0, r\omega) q(z, 0, r\omega). \end{aligned}$$

□

4.3 The central parameter and frozen convolution

For $\tau \neq 0$, set

$$(f \#_\tau g)(\eta) = (\pi|\tau|)^{-2n} \text{Os} \int e^{2i\omega(u,v)/\tau} f(\eta + u) g(\eta + v) du dv, \quad f \#_0 g = fg. \quad (4.7)$$

For full symbols a, b , write

$$(a \diamond b)(z, \tau, \eta) = (a(z, \tau, \cdot) \#_\tau b(z, \tau, \cdot))(\eta). \quad (4.8)$$

The changes $u = |\tau|^{1/2}U$, $v = |\tau|^{1/2}V$ reduce this product to $\#_{\text{sgn } \tau}$. For homogeneous symbols,

$$(p \star q)(z, \tau, \eta) = |\tau|^{(m+m')/2} (p_{\text{sgn } \tau} \#_{\text{sgn } \tau} q_{\text{sgn } \tau})(z, |\tau|^{-1/2}\eta), \quad \tau \neq 0. \quad (4.9)$$

Lemma 4.4 (The frozen product). *The product $\diamond$ maps $S_H^m \times S_H^{m'}$ to $S_H^{m+m'}$ continuously with respect to symbol seminorms and classical remainder estimates. If a, b have leading terms p, q , their product has leading term $p \star q$. When a, b are independent of the base variable, the full symbol of their composition in (2.7), without cutoffs, is exactly $a \diamond b$.*

Proof. First take Schwartz symbols. A left-invariant operator has the form $f \mapsto f \star k_a$, so the composed operator has convolution kernel $k_b \star k_a$, with the order dictated by (2.3). Fourier transformation in the central variable keeps the same parameter τ in both factors; the central term in the group law produces twisted convolution in the horizontal variables. A subsequent horizontal Fourier transform gives (4.7). At $\tau = 0$, this becomes pointwise multiplication. The sign can also be checked from

$$\eta_{x_j} \diamond \eta_{y_k} - \eta_{y_k} \diamond \eta_{x_j} = i\delta_{jk}\tau, \quad [D_{X_j}, D_{Y_k}] = i\delta_{jk}D_T. \quad (4.10)$$

For uniform estimates at the equator, put $R = \langle \sigma \rangle_H$ and split into the regions $|\tau| \geq cR^2$ or $|\eta| \geq c'R$. In the first region, normalize the horizontal variables by $|\tau|^{1/2}$ and apply Lemma 4.1 to symbol families with uniformly bounded seminorms. In the second region, put $h = \tau/R^2$ and normalize horizontally by R . If h stays away from 0, the same argument applies. If h is near 0, use stationary phase with parameter h in (4.7). The stationary point is $(u, v) = 0$, where the frequency arguments stay away from the zero section. Outside a neighborhood of that point, integration by parts gives remainders of every order in h . Regions where a frequency argument is near 0 are separated off; normalized seminorms grow only polynomially in the

small scale, and additional integrations by parts absorb that growth. The dyadic decomposition and moment cancellation in Lemma 5.1 give a uniform formulation of these estimates when $h \rightarrow 0$ and $R \rightarrow \infty$ simultaneously.

Central derivatives are controlled by the exact identity

$$\begin{aligned} \partial_\tau(a \diamond b) &= (\partial_\tau a) \diamond b + a \diamond (\partial_\tau b) \\ &\quad + \frac{i}{2} \sum_j ((\partial_{\eta_{x_j}} a) \diamond (\partial_{\eta_{y_j}} b) - (\partial_{\eta_{y_j}} a) \diamond (\partial_{\eta_{x_j}} b)). \end{aligned} \quad (4.11)$$

This follows by differentiating the twisted-convolution phase before Fourier transformation. Each term lowers order by 2; horizontal derivatives lower order by 1. We obtain (1.6) at order $m + m'$ with all derivatives. Applying these estimates to the remainders in (1.5), and using compatibility with dilations, gives a classical expansion for $a \diamond b$. Its leading term is (4.9), already glued smoothly in Theorem 4.3. Removing the Schwartz cutoffs using these estimates completes the proof. $\square$

Remark 4.5 (Two different expansions). In the isotropic calculus, B_k lowers order by $2k$. But $\tau^k B_k(a, b)$ does not lower Heisenberg order, because τ^k has order $2k$. Thus one must not take ab alone as the principal symbol of a Heisenberg composition. The entire exact product $\#_\pm$ contributes at principal order; the order reduction in the full calculus comes from variation in the base variable.

5 The composition formula and symbolic calculus

5.1 Weighted Taylor expansion and remainder estimates

Lemma 5.1 (Amplitude reduction). *Work in a relatively compact Darboux chart, with fixed cutoffs near the diagonal. Put $v = (v_0, v')$, $|\alpha|_H = 2\alpha_0 + |\alpha'|$, and $N \geq 1$. For an amplitude $F(z, v, \sigma)$, write $F \in \mathcal{A}^\mu$ if, on the fixed (z, v) -support,*

$$|\partial_z^\gamma \partial_v^\delta \partial_\sigma^\beta F| \leq C_{\gamma\delta\beta} \langle \sigma \rangle_H^{\mu - |\beta|_H} \quad \text{for all } \gamma, \delta, \beta. \quad (5.1)$$

Define

$$T_{<N} F(z, v, \sigma) = \sum_{|\alpha|_H < N} \frac{v^\alpha}{\alpha!} \partial_v^\alpha F(z, 0, \sigma).$$

Then the following assertions hold, modulo smooth kernels.

(i) **One amplitude.** If $F \in \mathcal{A}^\mu$, the kernel

$$K_F(z, v) = (2\pi)^{-d} \text{Os} \int e^{i\langle v, \sigma \rangle} F(z, v, \sigma) d\sigma$$

has a reduced symbol of the form

$$c_F = \sum_{|\alpha|_H < N} \frac{i^{|\alpha|}}{\alpha!} \partial_\sigma^\alpha \partial_v^\alpha F(z, 0, \sigma) + r_N, \quad |\partial_z^\gamma \partial_\sigma^\beta r_N| \leq C_{\gamma\beta N} \langle \sigma \rangle_H^{\mu - N - |\beta|_H}.$$

(ii) **Composed kernels.** Let a satisfy (1.6) of order m , and let $b(z, v, \sigma) \in \mathcal{A}^{m'}$. Set $k_b(z, v, u) = \mathcal{F}_{\sigma \rightarrow u}^{-1} b(z, v, \sigma)$ and

$$H(z, w) = \int k_a(z, v) k_b(z, v, wv^{-1}) dv.$$

Replace b by $T_{<N}b$ in this formula to obtain $H_{<N}$. The reduced symbol of $H - H_{<N}$ satisfies

$$|\partial_z^\gamma \partial_\sigma^\beta r_N(z, \sigma)| \leq C_{\gamma\beta N} \langle \sigma \rangle_H^{m+m'-N-|\beta|_H}. \quad (5.2)$$

In particular, this applies to $b(z, v, \sigma) = b_0(zv^{-1}, \sigma)$.

- (iii) **Classical symbols and phase changes.** If the amplitudes are classical, uniformly with all (z, v) -derivatives, then $r_N \in S_H^{\mu-N}$ in (i) and $r_N \in S_H^{m+m'-N}$ in (ii). For the phase changes in (2.9), a term $v^\alpha \sigma^\nu$ has order drop

$$|\alpha|_H - |\nu|_H; \quad |\alpha|_H - |\nu|_H \geq N \implies \text{a drop of at least } N.$$

Proof. (i) *One amplitude.* Ordinary Taylor expansion gives

$$F(v) = \sum_{|\alpha| < N} \frac{v^\alpha}{\alpha!} \partial_v^\alpha F(0) + N \sum_{|\alpha|=N} \frac{v^\alpha}{\alpha!} \int_0^1 (1-s)^{N-1} \partial_v^\alpha F(sv) ds.$$

The parameters (z, σ) are suppressed. Thus, with

$$I_N = \{\alpha : |\alpha| < N \leq |\alpha|_H\} \cup \{\alpha : |\alpha| = N\},$$

we have

$$F - T_{<N}F = \sum_{\alpha \in I_N} v^\alpha F_{N,\alpha}, \quad F_{N,\alpha} \in \mathcal{A}^\mu, \quad N \leq |\alpha|_H \leq 2N. \quad (5.3)$$

Indeed, the coefficients are respectively $\partial_v^\alpha F(0)/\alpha!$ and the integral coefficients above; (5.1) is preserved by both operations.

Integration by parts in σ yields

$$\text{Os} \int e^{i\langle v, \sigma \rangle} v^\alpha G d\sigma = i^{|\alpha|} \text{Os} \int e^{i\langle v, \sigma \rangle} \partial_\sigma^\alpha G d\sigma, \quad \partial_\sigma^\alpha G \in \mathcal{A}^{\mu-|\alpha|_H}.$$

For completeness, reduction preserves the order of any amplitude $G \in \mathcal{A}^s$: after a fixed compact cutoff in v ,

$$c_G(z, \zeta) = (2\pi)^{-d} \text{Os} \int e^{i\langle v, \vartheta \rangle} G(z, v, \zeta + \vartheta) dv d\vartheta.$$

Use $(1 - \Delta_v)^L e^{i\langle v, \vartheta \rangle} = \langle \vartheta \rangle^{2L} e^{i\langle v, \vartheta \rangle}$ and the anisotropic Peetre inequality. For $s_\beta = s - |\beta|_H$,

$$\begin{aligned} |\partial_z^\gamma \partial_\zeta^\beta c_G| &\leq C_L \int \langle \vartheta \rangle^{-2L} \langle \zeta + \vartheta \rangle_H^{s_\beta} d\vartheta \\ &\leq C'_L \langle \zeta \rangle_H^{s_\beta} \int \langle \vartheta \rangle^{-2L+|s_\beta|} d\vartheta \leq C''_L \langle \zeta \rangle_H^{s_\beta}, \quad 2L > d + |s_\beta|. \end{aligned}$$

Apply this to each coefficient in (5.3), after transferring v^α . Since $|\alpha|_H \geq N$, assertion (i) follows.

(ii) *Composed kernels.* Choose $1 = \sum_{j \geq 0} \varphi_j$, with

$$\text{supp } \varphi_j \subset \{\sigma : 2^{j-1} \leq R(\sigma) \leq 2^{j+1}\}, \quad j \geq 1.$$

For $a_j = \varphi_j a$ and $k_j = \mathcal{F}^{-1} a_j$, put

$$\tilde{a}_j(z, \zeta) = 2^{-jm} a_j(z, M_{2^j} \zeta), \quad k_j(z, \delta_{2^{-j}} V) = 2^{j(m+Q)} \mathcal{F}^{-1} \tilde{a}_j(z, V).$$

The $\tilde{a}_j$ have uniformly bounded derivatives on a fixed annulus. Fourier integration by parts therefore gives

$$|\partial_z^\gamma \partial_v^\beta k_j(z, v)| \leq C_{\gamma\beta L} 2^{j(m+Q+|\beta|_H)} (1 + |\delta_{2^j} v|)^{-L}. \quad (5.4)$$

In particular,

$$\|v^\alpha k_j(z, \cdot)\|_{L^1} \leq C_\alpha 2^{j(m-|\alpha|_H)}, \quad (\delta_{2^{-j}} V)^\alpha = 2^{-j|\alpha|_H} V^\alpha.$$

For $j \geq 1$, cancellation holds to every order:

$$\int v^\alpha k_j(z, v) dv = i^{|\alpha|} \partial_\sigma^\alpha a_j(z, 0) = 0. \quad (5.5)$$

These estimates also hold for $\mathcal{F}^{-1}(\varphi_k b)$, uniformly with all derivatives in its extra parameter v .

By (5.3), it suffices to estimate

$$H_{jk,\alpha}(z, w) = \int v^\alpha k_{a_j}(z, v) k_{b_{N,\alpha,k}}(z, v, wv^{-1}) dv, \quad \alpha \in I_N.$$

At comparable scales, $|j - k| \leq 2$, the substitutions $v = \delta_{2^{-j}} V$, $w = \delta_{2^{-j}} W$ give

$$\delta_{2^j}(wv^{-1}) = WV^{-1}, \quad dv = 2^{-jQ} dV,$$

and hence the order factor is

$$2^{j(m+Q-|\alpha|_H)} 2^{j(m'+Q)} 2^{-jQ} = 2^{j(m+m'-|\alpha|_H+Q)}.$$

At separated scales, subtract a Taylor polynomial from the factor paired with the smaller-scale kernel. The cancellation step is

$$\int k_j(v) G(v) dv = \int k_j(v) (G(v) - T_{<L} G(v)) dv, \quad j \geq 1.$$

For $j > k + 2$, each retained remainder factor contributes $2^{-L(j-k)}$; for $k > j + 2$, first use $u = wv^{-1}$ and apply the same identity to the k -kernel. Polynomial factors from the group law are absorbed by increasing L in (5.4).

Here is the resulting summable estimate. Write $c_{jk,\alpha} = \mathcal{F}_{w \rightarrow \sigma} H_{jk,\alpha}$ and $J = \max(j, k)$. For every L , after increasing the number of integrations by parts and Taylor terms,

$$|\partial_z^\gamma \partial_\sigma^\beta (\varphi_\ell c_{jk,\alpha})| \leq C_{\gamma\beta\alpha L} 2^{j(m-|\alpha|_H)+km'-\ell|\beta|_H} 2^{-L|j-k|} 2^{-L|\ell-J|}.$$

For $\ell > J$, the last factor follows from Fourier integration by parts. For $\ell < J$, expand the Fourier exponential and use (5.5); the coordinates of uv are polynomials in (u, v) , so their Taylor monomials are covered by the same cancellation. If a low-frequency block occurs, Taylor subtraction against the other block gives the bound instead. Fixed cutoffs contribute smooth kernels. Choosing $L > |m - |\alpha|_H| + |m'| + 2$ now gives

$$\sum_{j,k \geq 0} 2^{j(m-|\alpha|_H)+km'} 2^{-L|j-k|} 2^{-L|\ell-\max(j,k)|} \leq C_L 2^{\ell(m+m'-|\alpha|_H)}.$$

Sum over $\alpha \in I_N$ and use $|\alpha|_H \geq N$ to obtain (5.2). The estimates justify removal of frequency cutoffs and hold for every γ, β .

(iii) *Classical symbols and phase changes.* For classical amplitudes, apply the preceding estimates to their homogeneous terms and to remainders of arbitrarily low order. Homogeneous rescaling and Proposition 1.4 give

$$r_N \in S_H^{\mu-N} \quad \text{in (i)}, \quad r_N \in S_H^{m+m'-N} \quad \text{in (ii)}.$$

On the j -th annulus,

$$v^\alpha \sigma^\nu = 2^{-j(|\alpha|_H-|\nu|_H)} V^\alpha \zeta^\nu, \quad v = \delta_{2^{-j}} V, \quad \sigma = M_{2^j} \zeta.$$

After the leading linear change of variables in (2.9), every phase error has positive weight excess. Taylor's formula for its exponential, on the rescaled kernel, therefore has the same remainder bounds; its integral remainder is controlled by (5.4). This proves the phase assertion. $\square$

5.2 The full formula in left quantization

Theorem 5.2 (Composition formula). *If $A \in \Psi_H^m(X)$ and $B \in \Psi_H^{m'}(X)$, then $AB \in \Psi_H^{m+m'}(X)$ and*

$$\sigma_{m+m'}^H(AB) = \sigma_m^H(A) \star \sigma_{m'}^H(B). \quad (5.6)$$

In particular,

$$\sigma_{m+m'}^H(AB)(\pm) = \sigma_m^H(A)(\pm) \#_{\pm} \sigma_{m'}^H(B)(\pm). \quad (5.7)$$

If a, b are full left symbols in a Darboux chart, the full left symbol c of AB has the expansion, modulo Schwartz symbols,

$$c \sim \sum_{\alpha \in \mathbb{N}^{2n+1}} \frac{1}{\alpha!} (\partial_{\sigma}^{\alpha} a) \diamond (D_{\text{sym}}^{\alpha} b). \quad (5.8)$$

The term indexed by α has order $m + m' - |\alpha|_H$. The remainder after terms with $|\alpha|_H < N$ belongs to $S_H^{m+m'-N}$.

Proof. (i) The composed kernel. Work in one Darboux chart, suppressing cutoffs equal to 1 near the diagonal. By (2.7),

$$\begin{aligned} ABf(z) &= \int k_a(z, v)(Bf)(zv^{-1}) dv \\ &= \iint k_a(z, v)k_b(zv^{-1}, u)f(zv^{-1}u^{-1}) du dv. \end{aligned}$$

Since $v^{-1}u^{-1} = (uv)^{-1}$,

$$ABf(z) = \iint k_a(z, v)k_b(zv^{-1}, u)f(z(uv)^{-1}) du dv. \quad (5.9)$$

The calculations below may first be made with Schwartz frequency cutoffs; Lemma 5.1 justifies their removal.

(ii) Taylor expansion and symmetrization. Write $Z = (T, X_1, \dots, X_n, Y_1, \dots, Y_n)$ and set

$$V_v = \sum_{j=0}^{2n} v_j Z_j, \quad F(s) = k_b(z \exp(-sv), u), \quad F^{(k)}(s) = (-1)^k (V_v^k k_b)(z \exp(-sv), u).$$

For $|\alpha| = k$, define

$$Z_{\text{sym}}^{\alpha} = \frac{\alpha!}{k!} \sum_{\substack{j_1, \dots, j_k \in \{0, \dots, 2n\} \\ \#\{\nu: j_{\nu}=j\}=\alpha_j \text{ for every } j}} Z_{j_1} \cdots Z_{j_k}, \quad Z_{\text{sym}}^0 = \text{Id}.$$

Then

$$V_v^k = \sum_{|\alpha|=k} \frac{k!}{\alpha!} v^{\alpha} Z_{\text{sym}}^{\alpha}, \quad \frac{F^{(k)}(0)}{k!} = \sum_{|\alpha|=k} \frac{(-1)^k v^{\alpha}}{\alpha!} Z_{\text{sym}}^{\alpha} k_b(z, u).$$

For example, the coefficient of $v_j v_{\ell}$, $j \neq \ell$, at degree 2 is $(Z_j Z_{\ell} + Z_{\ell} Z_j)/2$, not $Z_j Z_{\ell}$. Taylor's formula with integral remainder gives

$$\begin{aligned} k_b(zv^{-1}, u) &= \sum_{|\alpha| < N} \frac{(-1)^{|\alpha|} v^{\alpha}}{\alpha!} Z_{\text{sym}}^{\alpha} k_b(z, u) + \mathcal{T}_N(z, v, u), \\ \mathcal{T}_N(z, v, u) &= \frac{(-1)^N}{(N-1)!} \int_0^1 (1-s)^{N-1} (V_v^N k_b)(z \exp(-sv), u) ds. \end{aligned}$$

Retaining only terms of weight $|\alpha|_H < N$ yields

$$k_b(zv^{-1}, u) = \sum_{|\alpha|_H < N} \frac{(-1)^{|\alpha|} v^\alpha}{\alpha!} Z_{\text{sym}}^\alpha k_b(z, u) + \mathcal{R}_N(z, v, u). \quad (5.10)$$

Here

$$\mathcal{R}_N = \mathcal{T}_N + \sum_{\substack{|\alpha| < N \\ |\alpha|_H \geq N}} \frac{(-1)^{|\alpha|} v^\alpha}{\alpha!} Z_{\text{sym}}^\alpha k_b(z, u).$$

Every monomial in $\mathcal{R}_N$ has weight at least N : $|\alpha| = N \Rightarrow |\alpha|_H \geq N$ in $\mathcal{T}_N$.

(iii) *Fourier transformation of each term.* For $t = uv$, put

$$h_\alpha(z, t) = \frac{(-1)^{|\alpha|}}{\alpha!} \int v^\alpha k_a(z, v) (Z_{\text{sym}}^\alpha k_b)(z, tv^{-1}) dv.$$

This is the frozen convolution in Lemma 4.4. Since

$$\mathcal{F}(v^\alpha k_a) = i^{|\alpha|} \partial_\sigma^\alpha a, \quad \mathcal{F}(Z_{\text{sym}}^\alpha k_b) = Z_{\text{sym}}^\alpha b, \quad D_{\text{sym}}^\alpha = i^{-|\alpha|} Z_{\text{sym}}^\alpha,$$

we obtain

$$\begin{aligned} \mathcal{F}_{t \rightarrow \sigma} h_\alpha &= \frac{(-1)^{|\alpha|} i^{|\alpha|}}{\alpha!} (\partial_\sigma^\alpha a) \diamond (Z_{\text{sym}}^\alpha b) \\ &= \frac{1}{\alpha!} (\partial_\sigma^\alpha a) \diamond (D_{\text{sym}}^\alpha b), \quad (-1)^{|\alpha|} i^{|\alpha|} = i^{-|\alpha|}. \end{aligned}$$

Moreover,

$$\partial_\sigma^\alpha a \in S_H^{m-|\alpha|_H}, \quad D_{\text{sym}}^\alpha b \in S_H^{m'}, \quad \mathcal{F} h_\alpha \in S_H^{m+m'-|\alpha|_H}.$$

(iv) *Remainder and principal symbol.* The remainder operator has the localized kernel expression

$$E_N f(z) = \iint k_a(z, v) \mathcal{R}_N(z, v, u) f(z(uv)^{-1}) du dv.$$

Lemma 5.1 gives its full symbol r_N with

$$r_N \in S_H^{m+m'-N}, \quad |\partial_z^\gamma \partial_\sigma^\beta r_N| \leq C_{N,\gamma,\beta} \langle \sigma \rangle_H^{m+m'-N-|\beta|_H}.$$

Thus the full symbol c of AB satisfies, modulo Schwartz symbols,

$$c = \sum_{|\alpha|_H < N} \frac{1}{\alpha!} (\partial_\sigma^\alpha a) \diamond (D_{\text{sym}}^\alpha b) + r_N, \quad N \geq 1.$$

Taking $N = 1$ and using Lemma 4.4,

$$\begin{aligned} c &= a \diamond b \pmod{S_H^{m+m'-1}}, & c &\in S_H^{m+m'}, & p &= \sigma_m^H(A), & q &= \sigma_{m'}^H(B). \\ \sigma_{m+m'}^H(AB) &= p \star q, & (p \star q)_\pm &= p_\pm \#_\pm q_\pm, \end{aligned}$$

Theorem 4.3 supplies smooth gluing at the equator. Proposition 2.3 preserves the principal term under changes of chart or quantization; a partition of unity gives the global operator. Terms supported away from the diagonal are smoothing by Lemma 2.2. $\square$

Remark 5.3 (The homogeneous terms). If $a \sim \sum_r a_{m-r}$ and $b \sim \sum_s b_{m'-s}$, then

$$c_{m+m'-\ell} = \sum_{r+s+|\alpha|_H=\ell} \frac{1}{\alpha!} (\partial_\sigma^\alpha a_{m-r}) \star (D_{\text{sym}}^\alpha b_{m'-s}). \quad (5.11)$$

The sum is finite at each order. For Weyl quantization, convert the full symbols to left symbols, apply this formula, and invert (2.8). Lower-order corrections depend on the quantization convention; the principal-symbol product $\star$ does not.

5.3 Borel summation and asymptotic completeness

Lemma 5.4 (Borel summation). *Given $a_j \in S_H^{m-j}$, there exists $a \in S_H^m$ such that*

$$a - \sum_{j < N} a_j \in S_H^{m-N} \quad \text{for every } N. \quad (5.12)$$

Among the symbols satisfying this condition, a is unique modulo Schwartz symbols. Similarly, given $A_j \in \Psi_H^{m-j}$, there exists $A \in \Psi_H^m$, unique modulo smoothing operators, such that $A - \sum_{j < N} A_j \in \Psi_H^{m-N}$.

Proof. Write $a_j = \rho^{-m+j} u_j$, with u_j smooth up to the boundary. Take $\chi \in C_c^\infty([0, \infty))$ equal to 1 near 0, and set

$$u = \sum_{j \geq 0} \chi(\rho/\varepsilon_j) \rho^j u_j.$$

Choose $\varepsilon_j \downarrow 0$ sufficiently rapidly that, for all sufficiently large j and all $k, J \leq j/3$,

$$\|\chi(\rho/\varepsilon_j) \rho^{j-J} u_j\|_{C^k} \leq 2^{-j}.$$

This is possible because k derivatives cost at most ε_j^{-k} , whereas the power contributes ε_j^{j-J} , and $j - J - k > 0$ for large j . For fixed k, J , the estimate holds for every sufficiently large j . Thus u is smooth, and its tail starting at $j = N$ is divisible by ρ^N in C^∞ . The finitely many terms where the cutoff differs from 1 vanish near the boundary. Taking $a = \rho^{-m} u$ gives (5.12); uniqueness follows from (1.7). For operators, quantize locally and patch using a partition of unity; Theorem 3.2 gives uniqueness. $\square$

Theorem 5.5 (Symbolic calculus). *The integer-order classes $\Psi_H^m(X)$ form an asymptotically complete filtered algebra with involution. More precisely:*

- (i) $\Psi_H^m \Psi_H^{m'} \subset \Psi_H^{m+m'}$, with formula (5.7).
- (ii) $A^* \in \Psi_H^m$ and $\sigma_m^H(A^*) = \overline{\sigma_m^H(A)}$.
- (iii) $\Psi^{-\infty}$ is a two-sided ideal and the intersection of all orders.
- (iv) Sequences of symbols and operators of decreasing order admit asymptotic sums.
- (v) The associated graded algebra is identified by

$$\bigoplus_{m \in \mathbb{Z}} \Psi_H^m / \Psi_H^{m-1} \simeq \bigoplus_{m \in \mathbb{Z}} (\mathcal{H}^m, \star).$$

Proof. Part (i) is Theorem 5.2. For Weyl quantization on half-densities, conjugating the kernel and exchanging its two variables gives the complex-conjugate symbol. Choose a real symmetric cutoff, or change the cutoff modulo smoothing. This proves (ii). If R is smoothing, regard it as an element of Ψ_H^{-N} for arbitrary N and apply (i) to AR, RA to obtain (iii). Part (iv) is Lemma 5.4. Part (v) follows from the exact sequence (3.4) and the composition formula. $\square$

6 Ellipticity and the construction of parametrices

6.1 Gluing the inverse symbol pair

Definition 6.1 (Heisenberg ellipticity). An operator $A \in \Psi_H^m(X)$ is *Heisenberg-elliptic* if, writing $p = \sigma_m^H(A)$, there are smooth sections $e_\pm \in S_{\text{iso}}^{-m}(\mathcal{P}_\pm)$ such that

$$p_\pm \#_\pm e_\pm = e_\pm \#_\pm p_\pm = 1. \quad (6.1)$$

Exact inverses in the symbol algebras are required, with uniform estimates after all base derivatives. Fiberwise invertibility alone, without smooth parameter dependence, is not the definition used here.

Lemma 6.2 (The inverses are matched). *If $p \in \mathcal{H}^m$ satisfies (6.1), the pair (e_+, e_-) automatically satisfies the matching condition at order $-m$. It glues to a unique $e \in \mathcal{H}^{-m}$ satisfying*

$$p \star e = e \star p = 1. \quad (6.2)$$

Proof. Let $p_0(\omega)$ be the common leading isotropic coefficient of $p_{\pm}$. At order 0, the inverse equation gives $p_0 e_{\pm,0} = 1$, so p_0 is nowhere zero and $e_{+,0} = e_{-,0} = p_0^{-1}$. Initially, the classical expansions of $e_{\pm}$ might contain the odd orders $-m-1, -m-3, \dots$. At order drop 1, the equation gives $p_0 e_{\pm,1} = 0$, so this coefficient vanishes. Inductively, all odd coefficients vanish: the expansions of $p_{\pm}$ contain only even order drops, and B_k lowers order by $2k$.

The unsigned even coefficients e_{ℓ} , extended homogeneously in η with degree $-m-2\ell$, are determined by

$$e_0 = p_0^{-1}, \quad e_{\ell} = -p_0^{-1} \sum_{\substack{a+b+k=\ell \\ b < \ell}} B_k(p_a, e_b), \quad \ell \geq 1. \quad (6.3)$$

These identities are restricted to $|\eta| = 1$ after homogeneous extension. Every term at level ℓ carries the sign $(\pm 1)^{\ell}$, so the expansions of $e_{\pm}$ are $\sum_{\ell \geq 0} (\pm 1)^{\ell} r^{-m-2\ell} e_{\ell}(\omega)$. Since $e_{\pm}$ are actual classical symbols, their remainders satisfy all required estimates. Proposition 3.5 gives the gluing, and injectivity of restriction proves (6.2). $\square$

Example 6.3 (Inverses modulo Schwartz symbols are insufficient). Put $s_0(\eta) = 2^n e^{-|\eta|^2}$. A Gaussian calculation in (4.1) gives

$$K_{q_+^W(s_0)}(q, q') = \pi^{-n/2} e^{-(|q|^2 + |q'|^2)/2} = g_0(q) \overline{g_0(q')}, \quad g_0(q) = \pi^{-n/4} e^{-|q|^2/2}.$$

Since $\|g_0\|_{L^2} = 1$, this is a rank-one projection, so $s_0 \#_+ s_0 = s_0$. The pair $p_+ = 1 - s_0$, $p_- = 1$ is matched at order 0, since the Schwartz function s_0 contributes no jets at the equator. It defines a homogeneous symbol $p \in \mathcal{H}^0$ in a contact trivialization. Both restrictions are invertible modulo Schwartz symbols, but p_+ is not exactly invertible: if $e \#_+ p_+ = 1$, multiplication on the right by s_0 would give $0 = s_0$. Thus even an equatorial symbol equal to 1 does not suffice for Heisenberg ellipticity. The Schwartz part on a hemisphere still contains essential information in the *Heisenberg principal symbol*; it is not a smoothing error on X .

6.2 Two-sided parametrices

Theorem 6.4 (Parametrix). *For $A \in \Psi_H^m(X)$, the following conditions are equivalent:*

- (i) *A is Heisenberg-elliptic.*
- (ii) *There exists $B \in \Psi_H^{-m}(X)$ such that*

$$AB - \text{Id} \in \Psi^{-\infty}(X), \quad BA - \text{Id} \in \Psi^{-\infty}(X). \quad (6.4)$$

The operator B is unique modulo smoothing operators, and $\sigma_m^H(B) = (\sigma_m^H(A))_{\star}^{-1}$.

Proof. Step 1: inversion at principal order. Lemma 6.2 gives $e \in \mathcal{H}^{-m}$ with $p \star e = e \star p = 1$. Surjectivity in (3.4) gives $B_0 \in \Psi_H^{-m}$ with principal symbol e . The composition formula and exactness give

$$R = \text{Id} - AB_0 \in \Psi_H^{-1}, \quad L = \text{Id} - B_0A \in \Psi_H^{-1}. \quad (6.5)$$

Step 2: removing the right error. The symbolic calculus gives $R^j \in \Psi_H^{-j}$. Take a Borel sum

$$S_R \sim \text{Id} + R + R^2 + \cdots \in \Psi_H^0, \quad B_R = B_0 S_R \in \Psi_H^{-m}.$$

No assumption $\|R\| < 1$ is needed: this is summation by order, not an operator-norm convergent Neumann series. For $S_R^{(N)} = \sum_{j=0}^{N-1} R^j$, we have

$$(\text{Id} - R)S_R^{(N)} = \text{Id} - R^N, \quad S_R - S_R^{(N)} \in \Psi_H^{-N}.$$

Hence $AB_R - \text{Id} \in \Psi_H^{-N}$ for every N , so it is smoothing.

Step 3: removing the left error and comparing. Similarly, set $S_L \sim \sum_{j \geq 0} L^j$ and $B_L = S_L B_0$. Then $B_L A - \text{Id}$ is smoothing. The identity

$$B_L - B_R = B_L(\text{Id} - AB_R) + (B_L A - \text{Id})B_R \quad (6.6)$$

shows that $B_L - B_R$ is smoothing. Since smoothing operators form a two-sided ideal, $B = B_R$ is also a left parametrix.

Step 4: uniqueness and the converse. If B, B' are both parametrices, then

$$B - B' = B(\text{Id} - AB') + (BA - \text{Id})B' \in \Psi^{-\infty}.$$

Conversely, taking principal symbols in (6.4) and restricting to $\mathcal{P}_\pm$ gives (6.1) with $e_\pm = \sigma_{-m}^H(B)(\pm)$. This also determines the principal symbol of B . $\square$

6.3 Full-symbol recursion and a consequence

To calculate a parametrix, use left quantization and write

$$a \sim \sum_{r \geq 0} a_{m-r}, \quad b \sim \sum_{s \geq 0} b_{-m-s}, \quad b_{-m} = e.$$

By (5.11), requiring the full symbol of the composition to equal 1 to every order gives, for $\ell \geq 1$,

$$b_{-m-\ell} = -e \star \sum_{\substack{r+s+|\alpha|_H=\ell \\ s < \ell}} \frac{1}{\alpha!} (\partial_\sigma^\alpha a_{m-r}) \star (D_{\text{sym}}^\alpha b_{-m-s}). \quad (6.7)$$

Every term on the right is known from earlier steps. Multiplication by e must occur on the left because $\star$ is noncommutative. Borel summation gives a right parametrix; Theorem 6.4 gives a two-sided parametrix modulo smoothing operators.

Corollary 6.5 (Hypoellipticity). *If A is Heisenberg-elliptic, u is a distribution, and Au is smooth, then u is smooth. The assertion also holds locally wherever a local parametrix exists.*

Proof. Write $BA = \text{Id} - K$, with K smoothing. Then $u = B(Au) + Ku$ is smooth by Lemma 2.2. For the local version, use nested cutoffs; terms with separated supports are smoothing. $\square$

References

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